PS-4



1. Title

Meeting-to-Execution Automation Agent

2. Background / Context

Meetings often result in important decisions, commitments, and follow-up actions. However, these outcomes are not always converted into clearly defined tasks with assigned owners and deadlines. Action items may be scattered across meeting notes, chat messages, and project management tools, making them difficult to track and manage.

An intelligent automation system can bridge the gap between discussions and execution by extracting actionable information from meeting transcripts or notes, organizing it into structured tasks, and routing it through an approval workflow before execution.

3. Problem Statement

Develop an AI-powered system that converts meeting transcripts or notes into structured decisions, action items, assigned owners, deadlines, and follow-ups. The system should distinguish confirmed commitments from suggestions and unresolved questions, identify duplicate or conflicting tasks, and support an approval workflow before assigning tasks or sending notifications.

The solution should help teams transform meeting outcomes into actionable, trackable tasks while reducing manual effort and preventing important commitments from being overlooked.

4. Objectives / Goals

- Extract decisions, action items, owners, and deadlines from meeting notes or transcripts.
- Distinguish confirmed commitments from suggestions and unresolved questions.
- Identify duplicate or conflicting tasks to improve task accuracy.
- Enable users to review and approve proposed tasks before execution.
- Integrate with a project management tool or export tasks in a structured format.
- Track task completion and highlight overdue or unassigned actions.
- Reduce the gap between meeting discussions and actual task execution.

5. Target Users / Stakeholders

- Project managers and team leaders
- Software development teams
- Corporate teams and business professionals
- Meeting coordinators and administrative teams
- Organizations managing multiple projects and follow-up activities

6. Constraints & Technical Considerations

- The system should support meeting transcripts or notes as input.
- Extracted tasks, owners, and deadlines should remain consistent with the provided meeting content.
- Confirmed commitments must be distinguished from suggestions and unresolved questions.
- Ambiguous task assignments should require user confirmation before execution.
- Notifications and task assignments should not be triggered without the required approval.
- Duplicate and conflicting tasks should be identified where possible.
- Integration with a project management tool or structured task export should be supported.
- The solution should remain feasible to implement and demonstrate within the hackathon duration.

7. Resources Provided

- **AMI Meeting Corpus:** A research dataset containing meeting recordings, transcripts, and annotations for studying meeting understanding and action-item extraction.
  Link: https://groups.inf.ed.ac.uk/ami/corpus/
- **QMSum Dataset:** A meeting summarization dataset containing meeting transcripts and query-based summaries, useful for testing meeting understanding and information extraction.
  Link: https://github.com/Yale-LILY/QMSum
- **Trello REST API:** Enables integration with Trello boards to create tasks, assign members, and manage due dates, subject to API access and permissions.
  Link: https://developer.atlassian.com/cloud/trello/rest/
- **GitHub REST API:** Supports task and issue management through GitHub repositories, including issue creation, assignment, and tracking.
  Link: https://docs.github.com/en/rest/issues

Teams may use their own meeting transcripts, notes, datasets, project management tools, or other relevant resources, provided they are suitable for the problem statement and comply with applicable access and licensing requirements.

8. Expected Solution / Key Features

The solution is expected to include:

- Meeting Input: Allow users to upload or provide meeting transcripts and notes.
- Decision and Task Extraction: Identify decisions, action items, owners, deadlines, and follow-ups.
- Commitment Classification: Separate confirmed commitments from suggestions and unresolved questions.
- Duplicate and Conflict Detection: Identify potentially duplicated tasks or conflicting commitments.
- Approval Workflow: Allow users to review, edit, approve, or reject proposed tasks before execution.
- Task Integration and Export: Send approved tasks to a project management tool or export them in a structured format.
- Confirmation Mechanism: Request clarification when task ownership or other important details are ambiguous.
- Task Tracking: Monitor task completion and highlight overdue or unassigned actions.
- Follow-up Management: Surface pending actions and follow-ups to improve accountability.

9. Expected Deliverables

Teams must submit:

- A functional application that converts meeting transcripts or notes into structured tasks.
- A task extraction component identifying decisions, action items, owners, deadlines, and follow-ups.

- A classification mechanism distinguishing commitments, suggestions, and unresolved questions.
- A duplicate and conflict detection component.
- An approval workflow for reviewing and confirming proposed tasks.
- A project management integration or structured task export mechanism.
- A demonstration showing transcript upload, task generation, approval, and transfer to a project board with owners and due dates.
- A brief technical explanation of the approach, workflow, and key implementation decisions.

10. Evaluation Criteria

- Accuracy of information extraction.
- Correct classification of commitments and suggestions.
- Effectiveness of duplicate and conflict detection.
- Reliability of approval and confirmation workflows.
- Effectiveness of task tracking and integration.
- Overall functionality and usability.